

HerHealth Matters: Maternal Health Intelligence Report

Strategic Alignment: UN Sustainable Development Goal 3 (Good Health & Well-being)

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1.0 Project Overview & Strategic Intent

A. The Problem Statement

Maternal mortality remains a critical global challenge, often driven by late-stage detection of physiological and psychological risks. In many clinical settings, maternal care is reactive rather than proactive, focusing on physical vitals while overlooking the complex, compounding factors—such as mental health and historical complications—that lead to high-risk outcomes. Without a unified, data-driven tool, healthcare workers are forced to make life-saving decisions without a complete picture of a patient's risk profile.

B. The Solution: HerHealth Matters Dashboard

The HerHealth Matters: Maternal Health Intelligence Dashboard is a predictive clinical solution designed to bridge this information gap. By integrating biometric data with AI-driven risk modeling, this project transforms raw patient records into an actionable "Triage Roadmap." It allows clinicians to move beyond simple monitoring to active prediction, identifying high-risk trajectories before they result in emergencies.

C. Global Impact: Alignment with SDG 3

This project is strategically engineered to support the United Nations 2030 Agenda for Sustainable Development, specifically focusing on:

- **SDG Target 3.1:** Reducing the global maternal mortality ratio by providing early-warning systems for high-risk pregnancies.
- **SDG Target 3.4:** Promoting mental health and well-being. Our core discovery—the 4.75x mental health risk multiplier—proves that psychological support is a clinical necessity for maternal survival.

2.0 Technical Framework: The Power BI Intelligence Engine

A. System Architecture & Design

The HerHealth Matters Maternal Health Dashboard was engineered using Microsoft Power BI, leveraging advanced data modelling, DAX (Data Analysis Expressions), and integrated Machine Learning. Unlike a static report, this is a dynamic "living" ecosystem designed to handle the complexities of maternal healthcare.

B. The Power of the Three-Page Integrated Flow

A single chart cannot tell the whole story of a patient. This dashboard utilizes a tri-modular architecture to ensure no risk factor is overlooked:

- **Page A (Population Overview):** Provides the "Macro" perspective for administrative resource planning and high-level risk tracking across 1,100 patients.
- **Page B (Clinical Analysis):** Shifts to the "Micro" perspective, allowing for biometric deep-dives and individual patient vitals monitoring at the point of care.
- **Page C (AI Insights):** Moves into "Predictive" territory, using machine learning to identify the invisible drivers (like the 4.75x Mental Health multiplier) that traditional analysis might miss.

C. Advanced Interactivity & Global Filtering

The core strength of this Power BI solution lies in its Cross-Report Interactivity:

- **Synced Slicers:** The dashboard features integrated "Slicers" for Age Group and Risk Level. When a nurse selects "High Risk" on Page A, the entire dashboard—including the clinical scatter plots on Page B and the AI trees on Page C—instantly reconfigures to show data *only* for that high-risk segment.
- **Drill-Down Capabilities: Every visual is interactive.** Clicking on a specific demographic or risk category automatically filters the surrounding charts, allowing healthcare workers to "drill down" from a population of 1,100 to a single vulnerable patient in seconds.
- **Tooltips & Tool-driven Narratives:** Hovering over data points reveals hidden metadata, providing clinicians with instant context without cluttering the visual interface.

Conclusion of Technical Intent

By combining descriptive, diagnostic, and predictive analytics into one synced Power BI environment, this project moves clinical decision-making from "guesswork" to "certainty." It empowers the healthcare team to visualize the pathway to risk before it becomes a crisis, directly supporting the mission of SDG 3.

3.0 Clinical Intelligence Walkthrough

Below is a detailed clinical walkthrough of the HerHealth Intelligence system, including interactive visual interpretations and utility guides for frontline healthcare workers.

3.1 Page A: HerHealth Overview (Population Pulse)

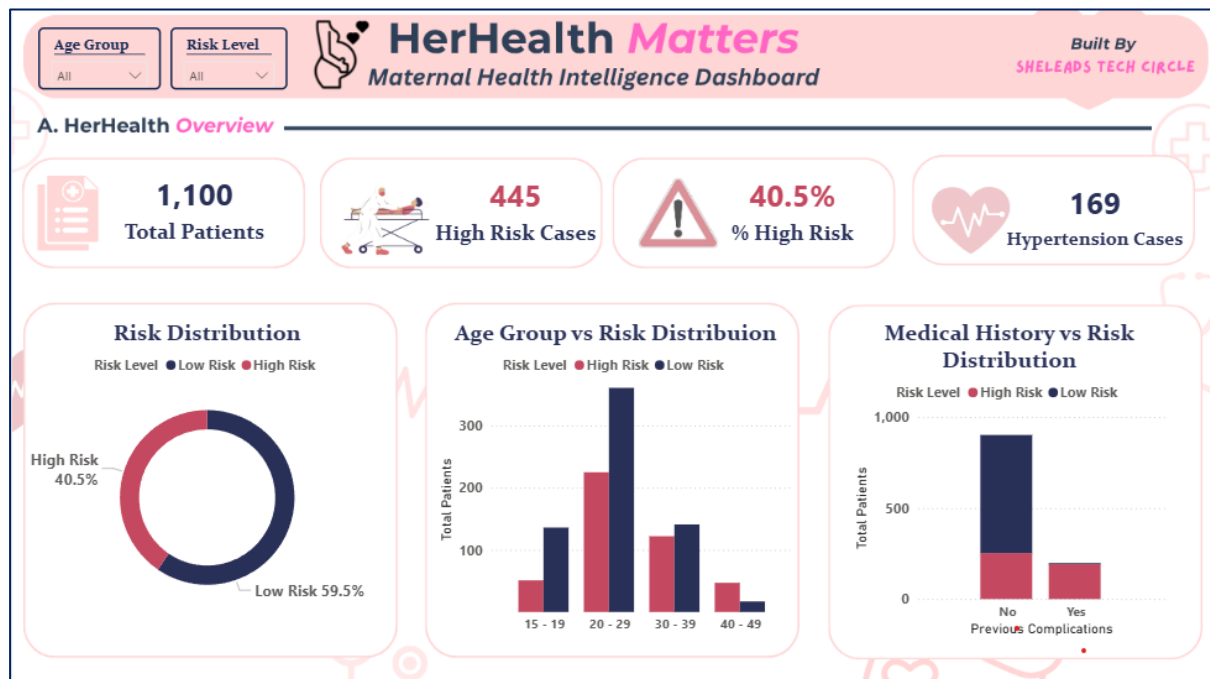


Figure 1: HerHealth Overview page

KPI & Visual Breakdown:

- **Total Patients (1,100):** This is the total sample size of the registry. It represents the scale of the clinic's current maternal responsibility.
- **High Risk Cases (445) & % High Risk (40.5%):** These are the most critical KPIs. They tell the healthcare worker that nearly **4 out of 10 patients** require intensive monitoring.
- **Hypertension Cases (169):** A specific count of patients with high blood pressure, identifying the immediate physiological burden on the clinic.
- **Age Group vs. Risk Distribution (Bar Chart):** This shows that the **20-29 age group** is the largest demographic. While it has many Low Risk cases, it also holds the highest volume of High Risk patients due to its size.
- **Medical History vs. Risk (Column Chart):** This visual proves a direct correlation: almost 100% of patients with **Previous Complications** (the "Yes" bar) fall into the High Risk category.

Healthcare Worker Utility:

- **Nurse Usage:** At the start of a shift, the nurse uses these KPIs to determine the "clinical load." If nearly half the patients are high-risk, the nurse knows to prepare the emergency kits and ensure the crash cart is accessible.

- **For the Triage Nurse:** Use this for **Daily Preparation**. By checking the Age Group chart, the nurse knows that the 20-29 cohort will require the most time and resources during a busy clinic day.

3.2 Page B: HerHealth Clinical Analysis (Biometric Deep-Dive)

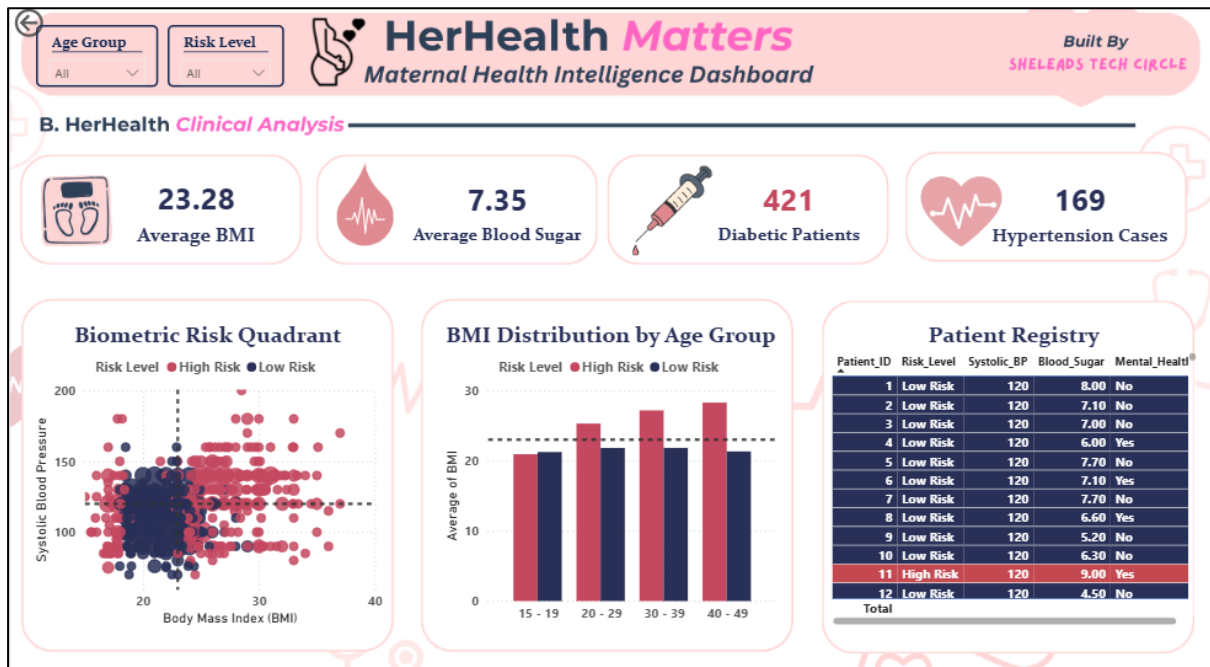


Figure 2: HerHealth Clinical Analysis

KPI & Visual Breakdown:

- **Average BMI (23.28):** Provides the baseline physical health of the population.
- **Average Blood Sugar (7.35) & Diabetic Patients (421):** These track metabolic health. With 421 patients being diabetic, the clinic knows that nearly **38% of patients** need nutritional and insulin monitoring.
- **Biometric Risk Quadrant (Scatter Plot):** This plots **Systolic BP vs. BMI**. The pink dots represent High Risk patients. You will notice they cluster at the top-right, where BP exceeds **135 mmHg**.
- **Patient Registry (Table):** A live list of every patient, their ID, and their specific vitals. This is the "Action List."

Healthcare Worker Utility:

- **For the Consulting Physician:** Use the **Biometric Risk Quadrant** as a **Diagnostic Shortcut**. During a check-up, if a patient's vitals land in that "Pink Cluster," the doctor can immediately escalate care without waiting for further symptoms.
- **Nurse Usage:** The **Patient Registry** is the nurse's digital "To-Do" list. She can filter the registry for "High Risk" and "Diabetic" to create a specific list of patients who need their blood sugar checked every hour. It ensures that no high-vulnerability patient is missed during a busy rotation.

- **Nurse Usage** (Hypertension cases): This allows the nurse to track the inventory of blood pressure medication. Knowing exactly how many hypertensive patients are in the registry helps in ensuring the clinic never runs out of essential medicine.

3.3 Page C: HerHealth AI Insights (The Discovery Engine)

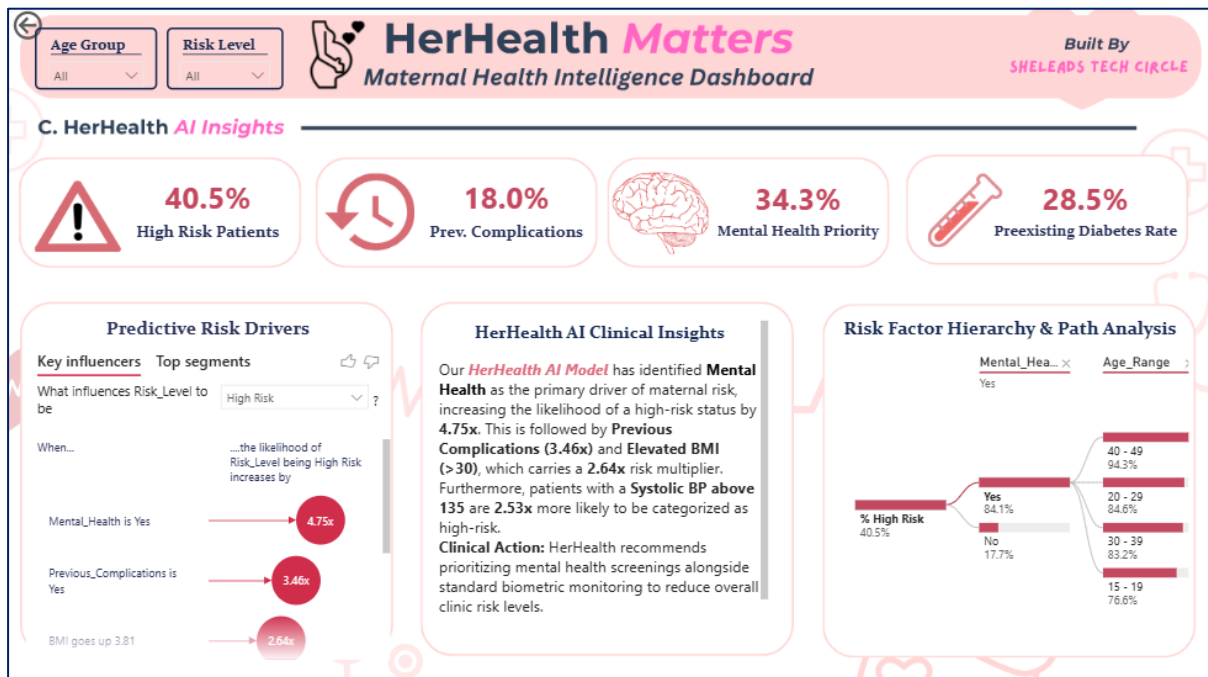


Figure 3: HerHealth AI Insights

KPI & Visual Breakdown:

- **High Risk Patients (40.5%):** This is the most critical KPI. They tell the healthcare worker that nearly **4 out of 10 patients** require intensive monitoring.
- **Mental Health Priority (34.3%):** Shows the percentage of the population identified as having mental health challenges.
- **Previous Complications (18.0%):** Shows the percentage of the population identified as having previous health complications.
- **Preexisting Diabetes Rate (28.5%):** Shows the percentage of the population identified as having had diabetes before conceiving.
- **Predictive Risk Drivers (Key Influencers):** This is our **4.75x Discovery using AI** (further explained on page 6: AI Methodology) . It shows that **Mental Health** is the strongest predictor of risk, far outweighing physical markers.
- **HerHealth AI Clinical Insights :** A summary based on the key influencers output that translates the math into plain English (e.g., "*Mental Health increases risk by 4.75x*").
- **Pathways to High Risk (Decomposition Tree):** This splits the data into "Risk Paths." It shows that for the **Mental Health "Yes"** group, the risk is 84.1%, but it climbs to **94.3%** for the **40-49 age group**.

Healthcare Worker Utility:

- **For the Midwife & Mental Health Counselor:** This visual proves to the board that **Mental Health is a Clinical Emergency**, not just a "wellness" issue. This discovery changes the **Intake Interview**. Knowing that **Mental Health carries a 4.75x risk multiplier**, the midwife will now spend more time asking about a patient's emotional well-being and stress levels, treating it with the same clinical importance as a blood pressure reading.
 - **For the Intake Nurse:** This is a **Triage Roadmap**. If a 45-year-old patient marks "Yes" for mental health struggles, the nurse can bypass standard triage and move her directly to the High Risk specialized unit, potentially saving her life.
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4.0 AI Methodology: *(Predictive Drivers explanation)*

A. Understanding the AI Core: Predictive Risk Drivers

The "**Predictive Risk Drivers**" section (utilizing the **Key Influencers** and **Top Segments** visuals) represents the machine learning core of the HerHealth Dashboard. Unlike standard charts that simply show "what happened," these AI components explain "**why it happened**" and "**what is likely to happen next**."

1. Key Influencers: The "Why" Engine

- **What it is:** This is a machine learning visual that uses a technique called **Logistic Regression** to analyze all 1,100 patients simultaneously. It looks for the specific factors that "push" a patient from Low Risk into High Risk.
- **How it works:** The AI compares every patient in the registry. It isolates one variable at a time (like BMI or Mental Health) while holding others constant to see which one has the strongest mathematical correlation with a "High Risk" outcome.
- **The Result:** This is how we discovered the **4.75x Mental Health multiplier**. The AI determined that out of all collected data, a "Yes" in the Mental Health category was the most statistically significant predictor of risk, followed by Previous Complications and Systolic BP.

2. Top Segments: The "Who" Engine

- **What it is:** While "Key Influencers" looks at single factors, **Top Segments** uses a **Decision Tree Algorithm** to find *combinations* of factors.
- **How it works:** The AI groups patients into "segments" or clusters. It might find, for example, that being aged 40-49 is a risk, but being aged 40-49 **AND** having a high BMI **AND** having Mental Health priorities creates a "Super-Segment" of risk.
- **The Result:** It identifies the highest-risk "pockets" in your population. It tells the clinic exactly which specific group of women needs the most urgent intervention based on their combined profile.

Why This AI Component is Crucial for Healthcare Workers

For a **Nurse or Midwife**, these AI components act as a "**Digital Consultant**."

- **Evidence-Based Triage:** Instead of relying on a "gut feeling," the nurse can point to the AI Influencer data to justify why a patient needs a priority referral. It provides a mathematical "second opinion."
- **Removing Bias:** AI looks only at the data. It helps ensure that every patient—regardless of their age or background—is assessed against the same rigorous predictive patterns.
- **Efficiency:** In a busy clinic, a midwife doesn't have time to cross-reference a patient's BMI against their mental health and their age. The **Top Segments** AI does this instantly, flagging the highest-risk combinations so the midwife can focus on care rather than data analysis.

5.0 Conclusion: Transforming Data into Maternal Equity

The **HerHealth Matters Dashboard** represents more than a technical milestone; it is a strategic intervention in the global effort to achieve **SDG 3: Good Health and Well-being**. By integrating a population of **1,100 patients** into a unified **Power BI Intelligence Engine**, we have moved beyond descriptive reporting into the realm of **Predictive Maternal Care**.

Key Strategic Takeaways:

- **The Power of AI:** Our discovery of the **4.75x Mental Health risk multiplier** provides a scientific mandate for clinics to integrate psychological screenings into standard prenatal protocols.
- **Operational Efficiency:** Through the **Clinical Biometric Quadrant** and **Interactive Patient Registry**, we have provided nurses and midwives with a "Point-of-Care" tool that reduces triage time and ensures immediate specialist referrals for high-risk cases (BP >135 mmHg).
- **Scalable Impact:** I have designed this framework to be modular. This system can be scaled across regional clinics and beyond, providing a standardized, data-driven approach to reducing maternal mortality.

The Path Forward

The data is clear: maternal risk is predictable, and therefore, maternal mortality is preventable. By leveraging **Machine Learning (Key Influencers)** and **Advanced Data Visualization**, the HerHealth Matters project provides the "Triage Roadmap" necessary to ensure that no mother is left behind. This dashboard is a testament to how technology, when aligned with the **Sustainable Development Goals**, can create a safer, more equitable future for women everywhere.